

CLASS 11TH BIOLOGY (SECTION A: BOTANY) ? CRUCIAL YEAR QUESTIONS

DIVERSITY OF LIFE

- Q1. Give detailed account of two and five kingdom classification. Their merits and demerits.
- Q2. Describe the general characters of kingdom Monera.
- Q3. Describe the general characters of Fungi and its classification.
- Q4. What are the distinguishing features between Viruses, viroids, and prions?
- Q5. State the characters of Mycoplasma and BGA.
- Q6. Define taxonomy, illustrate briefly binomial nomenclature, and explain the various rules of binomial nomenclature.
- Q7. Describe briefly Artificial, natural, and phylogenetic classification.
- Q8. Distinguish between Dinoflagellates and Diatoms.
- Q9. Name various asexual spores of Fungi and the steps of sexual reproduction.
- Q10. Define: Taxon, systematics, species, taxonomic hierarchy.
- Q11. What are slime moulds?
- Q12. What are the diagnostic features of Euglenoids?

PLANT KINGDOM

- Q13. Describe the important characteristics of gymnosperms, Pteridophytes, and their classification.
- Q14. Describe briefly the following classes: Chlorophyceae, Rhodophyceae & Liverworts.
- Q15. Write the general characters of Bryophytes.
- Q16. Describe briefly the economic importance of algae.
- Q17. Explain briefly the sexual reproduction in mosses.

MORPHOLOGY AND ANATOMY

- Q18. Give a brief account on the following terms:
(a) Phyllotaxy
(b) Placentation
(c) Inflorescence
(d) Aestivation
- Q19. What is a flower? Describe the parts of an angiospermic flower.
- Q20. Describe the various anatomical features of monocot and dicot stem.
- Q21. What are the morphological characteristic features of root and stem and write their different modifications.
- Q22. What are the anatomical differences between monocot & dicot root?

- Q23.** Differentiate between:
(a) Pinnately and Palmately compound leaf
(b) Racemose & Cymose Inflorescence
- Q24.** Illustrate the anatomical differences between monocot & dicot leaf with a diagram.
- Q25.** Describe various characteristics of Family Solanaceae with floral characters.
- Q26.** What is a pulvinus?
- Q27.** Give two examples of parietal & axile placentation.
- Q28.** Define ground tissue & stele.
- Q29.** What are casparian strips & where are they present?

PLANT PHYSIOLOGY

- Q30.** Explain the mechanism of the light reaction of photosynthesis. What are the various photosynthetic pigments?
- Q31.** Explain in detail the C3 cycle & C4 cycle.
- Q32.** Define respiration & explain various steps of glycolysis.
- Q33.** Describe the TCA cycle & define oxidative phosphorylation.
- Q34.** Write the physiological roles of auxin & gibberellin.
- Q35.** Explain briefly about photorespiration.
- Q36.** Write the differences between mesophyll cells and bundle sheath cells of C4 plants.
- Q37.** Explain the Chemiosmotic hypothesis.
- Q38.** What are the various complexes involved in the electron transport chain on the inner mitochondrial membrane?
- Q39.** Define respiratory quotient & write the RQ value of glucose, fats, and proteins.
- Q40.** Why is ABA known as the stress hormone? What are its physiological roles?
- Q41.** Write a short note on:
(a) Differentiation, De-differentiation & Re-differentiation
- Q42.** Discuss the physiological roles of cytokinin and ethylene.
- Q43.** What is plasticity? Define Heterophylly.
- Q44.** Define substrate-level phosphorylation.
- Q45.** What are the internal factors that affect photosynthesis?
- Q46.** Why is respiration known as an amphibolic pathway?
- Q47.** How many NADH and FADH₂ molecules are produced during Krebs cycle?
- Q48.** Name the precursor of auxin, gibberellin & ethylene.